MILITARY SPECIFICATION

RAILROAD TIE INSERTER AND EXTRACTOR, WHEEL MOUNTED, SELF-PROPELLED, DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, wheel mounted railroad tie inserter and extractor with salient features as specified herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting, and Starting.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.
FED-STD-595 - Colors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2230

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(Copies of standard required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupations Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J534 - Standard for Lubrication Fittings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The railroad tie inserter and extractor shall be a diesel-engine-driven, self-propelled unit, wheel mounted for 56-1/2 inch gage track, and complete with equipment necessary to handle, position, insert, and extract railroad ties. The tie inserter shall be hereafter referred to as the unit.

3.2 Standard commercial product. The unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish a unit for first article inspection and approval (see 4.2.1 and 6.4).

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3.4 Construction. The unit shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7 Safety. All parts subject to high operational temperatures, that are of such a nature or are so located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or fully guarded to the extent necessary to eliminate the hazard. The unit shall comply with OSHA regulations in effect at time of manufacture.

3.8 Threads. Screw threads for all thread securing or attaching devices and threaded parts shall be in accordance with FED-STD-H28. Machine threads shall be the standard national coarse or fine thread series. Pipe threads shall be the standard national taper pipe thread series.

3.9 Performance. The unit shall be operated by one man and shall lift one tie with the workhead assembly at the maximum reach position, align the prearranged tie for inserting, and insert the tie under the rails to the correct resting position with minimum assistance from the gang crew. Time required to insert three ties shall be 1 minute (nominal). The unit shall be capable of self-propulsion to speeds of 20 miles per hour (mph), deadheading travel speed, 4 percent gradeability.

3.10 Powerplant. The engine shall be of the compression ignition type, either two or four-stroke cycle, with not less than three cylinders, capable of operating on diesel fuel as recommended by the manufacturer. The engine shall be equipped with an adequate and efficient fuel injection mechanism, adequate oil-filter system, efficient lubricating oil system, air cleaner of the oil-bath or equivalent dry-filter type and exhaust system. The engine shall be furnished with the manufacturer's standard accessories and, as a minimum, compete with the following accessories:

a. An instrument and control panel located at the operator station fitted with, but not limited to, the following:

(1) Start-stop engine control.
(2) Ammeter, charging indicator or voltmeter.

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(3) Liquid quantity fuel indicator.
(4) Oil pressure gage or indicator.
(5) Engine temperature gage or indicator.
(6) Tachometer.
(7) An engine hour meter having a totalizing mechanism of not less than 9999 hours shall be furnished. The meter shall accurately record the engine operating time and provide trouble-free performance under the operating conditions indicated in this specification.

3.11 Powerplant systems.

3.11.1 Starting. Engine shall start in any ambient temperature above 20 degree Fahrenheit (°F). The unit shall have a 12 volt electrical cranking system. Starting aids may be either electric glow plug or ether primer. When an ether priming system is furnished, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces.

3.11.2 Fuel. Fuel tank shall have sufficient capacity for not less than 8 hours of normal operation. A fuel shutoff valve attached directly to the tank shall be provided. A fuel gage located at the operator's station shall be furnished.

3.12 Electrical system.

3.12.1 Electrical. The unit shall have a 12-volt electrical system. An alternator having not less than a 37-ampere rating shall be furnished. All wiring shall be protected against weathering and wearing, and shall be supported or secured at intervals not exceeding 2 feet.

3.12.2 Battery. The unit shall be equipped with a 12 V battery. The battery shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

3.13 Drive. A variable speed hydraulic drive mechanism shall be provided to propel the unit forward and rearward to a maximum top speed of not less than 20 mph. The drive shall provide torque characteristics necessary to negotiate grades normally encountered in railroad maintenance. Drive wheels shall be of cast steel type and not less than 16 inches in diameter.

3.14 Brakes. The manufacturer's standard service brake system shall be provided complete with parking and emergency controls. Parking brake shall securely hold the unit from rolling on grades normally encountered in railroad maintenance.

3.15 Hydraulic system.

3.15.1 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be provided with a filler opening, a removable oil strainer, and a vented filler cap, accessibly located to facilitate filling the tank. The vented filler cap shall have means to preclude fluid splashing out of tank during movement of the unit. Provisions shall be made for draining of the reservoir. The reservoir drain shall be so located that the fluid will not drain on other components of the unit. A sight level or dip stick gage shall be furnished to determine level of fluid.

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3.15.2 Hydraulic oil filter. Manufacturer's standard hydraulic oil filtration system shall be furnished.

3.15.3 Hydraulic pump(s) system. A hydraulic pump(s) with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The hydraulic pump shall be driven by a heavy-duty power takeoff unit. When applicable, chain drivers shall be permitted. The pumping system shall include an adjustable pressure relief valve or unloading valve with the necessary range of adjustment.

3.15.4 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders.

3.15.5 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failure and maximum protection from damage during operations of the workhead assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered, and where installation of components requires a flexible connection. Hydraulic line and fittings shall be installed to preclude vibration and chafing. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently.

3.16 Workhead assembly. The manufacturer's standard hydraulically actuated articulating or telescoping section shall be provided complete with tie handling device. Under dynamic conditions, the gripper head assembly shall grip not less than 1,000 pounds placed on the tie.

3.16.1 Horizontal reach. With the workhead assembly in its most horizontal position, a minimal horizontal reach of not less than 7 feet. The reach shall be measured from the center line of the track.

3.16.2 Turntable operation. When specified (see 6.2), the turntable or swing assembly shall be hydraulically actuated by separate mechanisms and shall operate independently or in any combination of raising, lowering, and turning at the same time. When in combination, hand metering of the controls may be necessary. Turn movements shall be smooth with no lurching or binding. Turntable shall swing a total of not less than 180° to allow tie insertion from either side of the track. Tie insertion force shall be not less than 8,500 pounds per square inch.

3.16.3 Controls. A complete set of hand-operated controls shall be provided to actuate the unit to handle and insert ties. The controls shall be self-centering and shall revert to a neutral position when released by the operator. Placement of the controls shall be so as to prevent inadvertent actuation by the operator. The controls shall allow the operator to lift, align, and insert ties in less than 20 seconds per cycle.

3.16.4 Tie handling. For normal storage and travel of the unit, the tie handling section shall retract to prevent damage during transit.

3.17 Setoff device. Manufacturer's standard setoff device shall be provided.

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3.18 Covered operator's station. The covered operator's station shall be equipped with seat, instrument panel, and operating controls. The seat shall be padded and covered with weatherproof material. Instruments, controls, and switches specified in 3.10 and 3.16.3 shall be located on the instrument panel. All hand and foot operated controls including brake, throttle, transmission, hydraulic, and steering shall be within easy reach of the operator while sitting in his normal operating position. Instruments, gages, switches, and controls shall be protected against moisture and inclement weather conditions. Each instrument, gage, switch, and control shall be provided with permanently affixed identification markings.

3.19 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall be No. 13538, yellow, conforming to FED STD No. 595.

3.20 Lubrication. Lubrication shall be in accordance with the manufacturer's recommendation. Pressure lubrication shall not damage seals or other parts. All parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used. Hydraulic lubrication fittings shall conform to SAE J534.

3.21 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.22 Identification plate. An identification plate will be furnished by the contracting officer for each unit. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each unit in a conspicuous place with brass screws or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.23 Workmanship.

3.23.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.23.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.23.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.23.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one unit when a first article is required (see 3.3 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1, and the packaging inspection of 4.5.

4.3 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Operational test. The operational test for the unit shall cover all controls, electrical, hydraulic, engine, brake and drive systems. The unit shall be driven at all speeds, brakes tested for dependability, and all controls, mechanisms, and movements tested. Nonconformance to applicable requirements of the specification shall constitute failure of this test.

4.4.2 Performance tests. The unit shall be subjected to the following tests. Failure to pass any phase of these tests shall be cause for rejection. The tests shall, as a minimum, assure that the unit is capable of meeting the performance requirements as specified in section 3. The unit shall be operated sufficiently to verify the proper functioning of components. The workhead assembly shall be raised and articulated to the most horizontal position to verify length requirement of 3.16.1. The workhead assembly shall operate in all positions without evidence of binding or lurching. One tie lifted from the rail tops shall be aligned and inserted by the unit in conformance with paragraph 3.16.1. The unit shall be propelled to maximum speed to verify conformance to 3.13.

4.5 Packaging inspection. The unit shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Level C. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The equipment shall be lubricated for service in accordance with the technical manual. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

6. NOTES

6.1 Intended use. The railroad tie inserter and extractor unit covered by this specification is intended for use at established Naval Stations for inserting and extracting of railroad ties.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.

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(b) When specification shall be other than as specified (see 2.1.1).
(c) When a first article is required for inspection and approval (see 3.3, 4.2.1, and 6.4).
(d) When turntable is supplied (see 3.16.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete tractor or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

Custodian:

Navy - YD

Preparing activity:

Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-29230(YD)	RAILROAD TIE INSERTER AND EXTRACTOR, WHEEL MOUNTED, SELF-PROPELLED, DIESEL-ENGINE-DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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